TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Compound semiconductor nonlinear photonics for efficient lightwave generation and conversion			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum (√) (Please choose at least one of the areas above)			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical・Electronics		Electrical・Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Optical Application Devices	Optical Devices	Semiconductor Devices and Systems	Compound Semiconductor Devices
Concept of Proposed Technology	The proposed technology is built on the integrated III-V compound semiconductor nonlinear photonics platform, in particular with InGaP, that exhibits a state-of-the-art nonlinearity-to-loss ratio among all integrated photonic materials. In contrast to most nonlinear photonics platforms, InGaP combines a large intrinsic $\chi^{(2)}$ coefficient with a wide electronic bandgap, leading to low losses from visible-red to telecommunication wavelengths. This enables strong nonlinear interactions at optical intensities compatible with semiconductor lasers and dense photonic integration. Through phase-matched waveguides and high-Q microresonators, the platform supports highly efficient second-harmonic generation, broadband sum- and difference-frequency generation, and low-threshold optical parametric oscillation (OPO). These $\chi^{(2)}$-based processes enable parametric gain and coherent light generation on chip at milliwatt-level pump powers, a regime that has been difficult to access with existing integrated photonic platforms. It also enables ultra-efficient integrated entangled photon sources, which is critical for quantum networking applications.			

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	Critically, this technology enables compact, energy-efficient, and wavelength-agile photonic functions, including chip-scale parametric light sources and flexible wavelength translation across the visible, near-infrared, and mid-infrared spectral regions. The InGaP platform is compatible with wafer-scale fabrication and heterogeneous integration, making it suitable for scalable manufacturing and industrial deployment. Key Enabling Applications: • Optical communications: on-chip wavelength conversion, tunable transmitters, and band-to-band translation for next-generation telecom and datacom transceivers. • Sensing and spectroscopy: compact broadband and tunable light sources for chemical, environmental, and biomedical sensing systems. • Advanced photonic systems: low-noise optical parametric oscillators and frequency-agile sources for LiDAR, metrology, and microwave–photonics applications. • Quantum technologies: quantum frequency conversion, nonclassical light generation, and interfaces between heterogeneous quantum systems.
Rationale for Joint R&D Project	While InGaP $\chi^{(2)}$ nonlinear photonics has reached world-leading performance at the device level, its broader impact depends on successful translation into manufacturable and application-specific subsystems. This transition requires expertise in laser integration, electronic–photonic co-design, advanced packaging, reliability qualification, and cost optimization—capabilities that extend beyond academic research. Korean SMEs are globally competitive in compound-semiconductor processing, laser technology, electronics integration, and photonic module manufacturing. A joint R&D effort enables tight co-optimization between nonlinear photonic devices and system-level constraints, de-risks technology transfer, and accelerates the path toward commercialization. Without SME participation, these devices remain laboratory demonstrations; with SME participation, they become deployable photonic subsystems.
Keywords	1. Compound semiconductor photonics 2. Integrated parametric light sources 3. Wavelength conversion 4. Entangled photon sources

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3. Research and Development Objectives & Plan

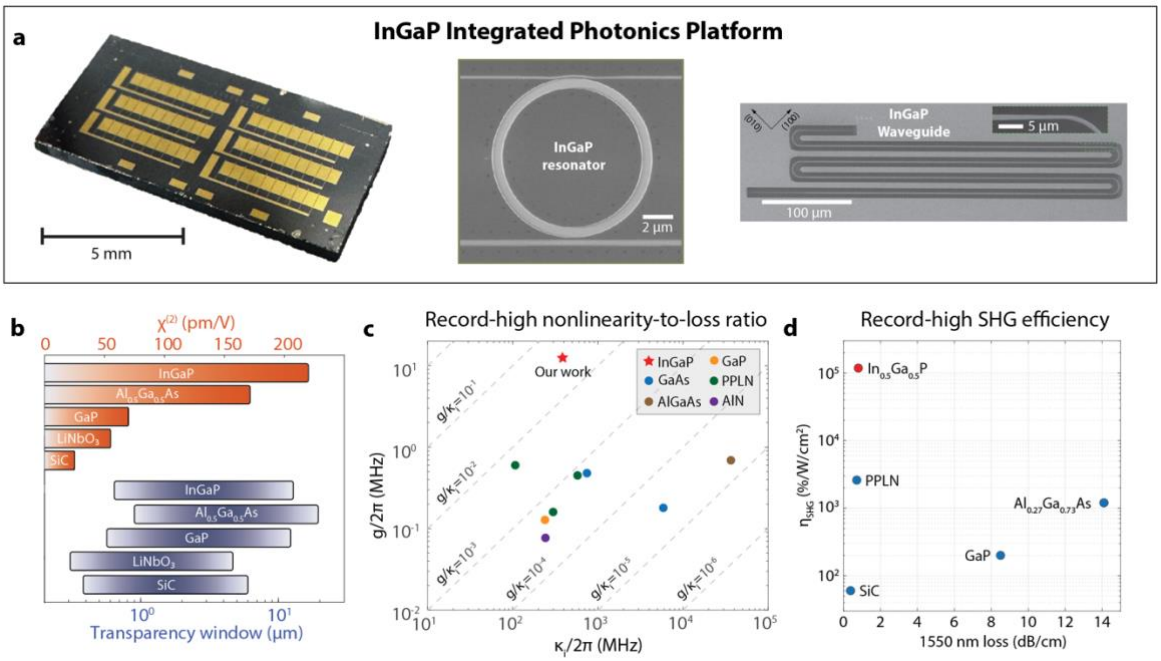
Technology Readiness Level (TRL)[2]	Start of the Project	Level (3-4)	End of the Project	Level (5-6)

R&D Objectives

My group at UIUC has pioneered the epitaxial InGaP $\chi^{(2)}$ integrated nonlinear photonics platform, demonstrating the highest nonlinearity and highest nonlinearity-to-loss ratio among all integrated photonics platforms [1,2,3]. We realized ultra-efficient second-harmonic generation in both InGaP microresonators and waveguides with efficiencies one to two orders of magnitude higher than the state of the art [1,2]. We also realized quantum frequency conversion between visible and telecom bands [4] as well as entangled photon generation with record-high efficiency using the InGaP platform [1,2].

The primary objective of this project is to transition InGaP $\chi^{(2)}$ nonlinear photonics from high-performance laboratory demonstrations to robust, application-ready platforms. Specific objectives include:

- Standardizing low-loss InGaP waveguide and resonator fabrication suitable for volume manufacturing.
- Demonstrating ultra-low-threshold optical parametric oscillators with stable, low-noise operation.
- Enabling broadband and efficient wavelength conversion tailored to target application bands.
- De-risking integration with semiconductor laser sources, electronic control circuits, and packaging solutions.
- Establishing design and performance guidelines that support industrial adoption.



R&D Target Specifications

- OPO threshold < 1 mW on-chip pump power
- Nonlinear conversion efficiency > 100,000%/W·cm² with > 1-cm-long waveguides
- Frequency conversion bandwidth > 10 nm and tunable cross visible to near-infrared.
- Entangled photon sources with rate > 1 THz/mW.
- High spectral coherence and long-term operational stability
- Wafer-scale reproducibility and yield suitable for pilot production

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R&D Plan	Year 1: Device and system co-design. Optimize waveguide dispersion, resonator geometries, and phase matching for $\chi^{(2)}$ nonlinear processes. Define application-specific specifications jointly with SME partners. Year 2: Fabrication and validation. Fabricate prototype devices and evaluate nonlinear efficiency, OPO threshold, thermal stability, noise performance, and yield. Iterate designs to address manufacturability. Year 3: Integration and demonstration. Integrate nonlinear devices with laser sources and packaging solutions. Demonstrate system-level operation in representative use cases such as wavelength conversion, parametric light and entangled photon generation.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	UIUC will lead innovation in $\chi^{(2)}$ nonlinear photonic device physics and technology. Responsibilities include photonic device modeling and design, nanofabrication, and advanced optical characterization. UIUC will also coordinate system architecture, performance benchmarking, and knowledge transfer.
Expected R&R of Korean SME Partner	The Korean SME partner will focus on industrialization and application alignment. Responsibilities include, for example, electronic–photonic packaging, environmental and reliability testing, pilot-line manufacturing, and commercialization planning. The SME partner will guide cost, yield, and market requirements.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	Near-term outcomes include prototype integrated frequency converters, optical parametric oscillators, turn-key quantum light sources that significantly outperform existing solutions in efficiency, size, and power consumption. These prototypes will demonstrate clear advantages for integrated photonic systems requiring wavelength agility and low energy use. In the medium term, the technology will enable differentiated photonic subsystems for optical communications, sensing, and quantum networking. In the long term, InGaP χ^2 nonlinear photonics provides a foundational technology for emerging quantum and hybrid photonic systems. The joint R&D effort will strengthen Korea's compound-semiconductor and photonics ecosystem and create strong opportunities for global commercialization.

6. References

[1] M. Zhao and K. Fang, "InGaP quantum nanophotonic integrated circuits with 1.5% nonlinearity-to-loss ratio", Optica 9, 258-263 (2022).
[2] J. Akin, Y. Zhao, Y. Misra, N. Haque, and K. Fang, "InGaP $\chi^{(2)}$ integrated photonics platform for broadband, ultra-efficient nonlinear conversion and entangled photon generation", Light: Science & Applications 13:290 (2024).
[3] J. Akin, Y. Zhao, A.K.M.N. Haque, and K. Fang, "Perspectives on epitaxial InGaP for quantum and nonlinear optics", Applied Physics Letters 125, 260501 (2024).
[4] J. Hu, H. Yuan, J. Akin, A.K.M.N. Haque, Y. Zhao and K. Fang, "Efficient bidirectional quantum frequency conversion between telecom and visible bands using programmable III-V nanophotonic waveguides", arXiv:2510.16696 (2025).

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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